Ecological Economics 204 (2023) 107645

Contents lists available at ScienceDirect

Ecological Economics

journal homepage: www.elsevier.com/locate/ecolecon

Methodological and Ideological Options

An index to measure the sustainability of place-based development pathways

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ARTICLE INFO

Keywords:
Sustainability
Transitions
Tropical Forests
Local Development
Amazon
Infrastructure

ABSTRACT

One of the main goals of modern sustainability science is to generate knowledge that societies can use to move toward more sustainable development pathways; however, there are few quantitative methods available to assess such pathways. This paper proposes a new index (the Sustainable Development Pathway Index, SDPI) that uses the Euclidean norm to measure the distance between the ideal and the actual development pathways of any geopolitical unit in a determined time frame. We test the new index by evaluating the development pathways of the 517 municipalities of the Brazilian Amazon, one of the world's last economic frontiers, from 1991 to 2010, by simultaneously analyzing indicators of ecological and socioeconomic infrastructures. We show that the region can be described as a mosaic of geopolitical units following distinct development pathways in response to the economic activities and extensive land transformations advancing from the region's southern and eastern borders to its center. This general pattern aligns well with the region's recent history. We outline a research agenda for using the SDPI to document the broad patterns of development pathways and the processes that make them sustainable or unsustainable.

1. Introduction

Human development is a political process by which societies make collective decisions about using their assets to improve their members' well-being. Such assets compose societies' infrastructure, which encompasses all the elements of connected systems that supply goods and services essential to enabling, sustaining, or enhancing societal well-being (Frischmann, 2013; Silva and Prasad, 2019; Thacker et al., 2019). There are two main types of infrastructure: ecological (or green) and socioeconomic (or grey). Ecological infrastructure (hereafter, "EI") is a network of natural, semi-natural, and restored areas designed and managed at different spatial scales to maintain biodiversity and provide the ecosystem services a society depends upon (Benedict and McMahon, 2006; Silva and Wheeler, 2017). It represents the stocks of renewable natural capital (Barbier, 2016; Costanza and Daly, 1992; Helm, 2019) a society has available in its territory as well as encompasses all types of ecosystems (marine, freshwater, and terrestrial) and settings (urban and rural) (Maes et al., 2015; Silva and Wheeler, 2017). On the other hand, socioeconomic infrastructure (hereafter, "SEI") combines all human-made assets or capitals (manufactured, human, and social; see Costanza and Daly, 1992) used by the social sectors (e.g., education, healthcare, culture) and economic sectors (e.g., finances, agriculture, energy, water) to deliver essential human-made services required for improving collective well-being (Silva and Prasad, 2019; Silva and Topf, 2020).

Because societies form a vast network of interacting and complex socio-ecological systems (Chapin et al., 2009), their asset stocks (EI and SEI) constantly change. These changes over time determine a society's development pathway (Barbier, 2016; Collados and Duane, 1999). Development pathways can be either sustainable or unsustainable, depending on the concept of sustainability used and how societies manage the trade-offs and synergies between the two infrastructure types. While the weak sustainability concept assumes that EI and SEI are totally substitutable, the strong sustainability concept assumes that the EI and SEI are complements rather than substitutes (Daly, 2020; Ekins et al., 2003). Because scientific and ethical grounds do not support the principles of weak sustainability, strong sustainability is the preferred starting point for any analysis of development pathways (Biely et al., 2018; Costanza and Daly, 1992; Ekins et al., 2003). Strong sustainability does not require that all natural ecosystems be kept as they are (Barbier,

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https://doi.org/10.1016/j.ecolecon.2022.107645
Received 10 June 2022; Received in revised form 15 August 2022; Accepted 12 October 2022
Available online 27 October 2022

2019; Costanza and Daly, 1992). Still, at a minimum, sustainable societies must design and implement EIs large and comprehensive enough to conserve their critical natural capital (hereafter, "CNC"). CNC represents the stocks of native or restored ecosystems responsible for important environmental functions (regulation, production, habitat, and information) that cannot be substituted or provided by human-made infrastructure (Ekins et al., 2003). The proportion of conserved land and sea needed to maintain a society's CNC is expected to vary geographically because species and ecosystems are not evenly distributed worldwide (Jenkins et al., 2013; Ladle and Whittaker, 2011; Smith et al., 2018).

Human development is usually assessed using composite indices, such as the Human Development Index (HDI), which is the geometric mean of indicators of income, education, and health (UNDP, 2020). However, HDI considers only a few outcomes of the services provided by SEI; thus, it alone cannot be used to distinguish between sustainable and unsustainable development pathways. Several approaches have been proposed in the quest to find more reliable indicators to examine the sustainability of human development. One approach adds indicators of human pressures on the environment (such as per capita CO2 emissions or material footprint) to the three original HDI indicators (Bravo, 2014). A second approach uses discount formulas to adjust the HDI to human environmental impacts (e.g., de la Vega and Urrutia, 2001; UNDP, 2020; Zhang and Zhu, 2022). A third approach focuses on assessing the efficiency of human development by using only socioeconomic indicators (Assa, 2021) or dividing HDI or one of its components (e.g., longevity) by one or more indicators of human pressures on the environment (Dietz et al., 2012; Dietz and Jorgenson, 2014; Givens, 2015; Hickel, 2020; Knight and Rosa, 2011). A fourth approach suggests that development pathways can be assessed by continuously monitoring a society's inclusive wealth, which is an aggregate estimate of the value of all the assets of a society (Polasky et al., 2015). Finally, a fifth approach rejects the idea of using a single composite index altogether and suggests representing human development as dashboards that include indicators of well-being, SEI, and EI (OECD, 2011). The assumptions of all these approaches have been challenged (e.g., Assa, 2021; Comim and Hirai, 2022; Roman and Thiry, 2016), and none of them alone responds to two of the most fundamental questions related to the sustainability of development pathways: (a) what does a sustainable society look like? and (b) how can sustainable and unsustainable development pathways be set apart?

An approach that has the potential to respond to these two questions simultaneously was proposed by Collados and Duane (1999) and modified by Silva and Topf (2020) when evaluating the sustainability of regional (or place-based) development pathways. They suggest that because a society's quality of life depends on ecosystem services (provided by EI) and human-made services (provided by SEI), its development pathway can be studied by graphically analyzing the successive changes in the indicators of these two types of services in a bi-dimensional Cartesian plane. Although innovative, this approach has never been evaluated empirically because quantitative methods for comparing vectors representing distinct development pathways were unavailable. In this paper, we close this gap by proposing and testing the Sustainable Development Pathway Index (SDPI). This new method uses the Euclidean norm to measure the distance between the ideal sustainable development pathway and the actual development pathway of any geopolitical unit. In addition, we use the SDPI to evaluate the development pathways of the 517 municipalities of the Brazilian Amazon, one of the world's last economic frontiers, from 1991 to 2010, and discuss the main findings of this test. We chose the Brazilian Amazon because its size and vast environmental and social heterogeneity make it a unique case study to examine how local factors shape development pathways in large regions.

2. Material and methods

2.1. The Sustainable Development Pathway Index (SDPI)

The theoretical framework proposed by Collados and Duane (1999) and modified by Silva and Topf (2020) suggests that human well-being (a state in which a person can meet all basic needs, pursue individual goals, thrive in society, and feel satisfied with life) is a function of the ecosystem and human-made services provided by EI and SEI, respectively. Because of this, societies (usually represented by geopolitical units with at least some degree of political autonomy) pursuing sustainable development pathways must simultaneously conserve their CNCs and maximize the efficiency of their SEI to produce and maintain gains in well-being. Under this framework, geopolitical units (ranging from local to national) are represented as points in a bi-dimensional ($\mathbb{R}^2$) Cartesian plane in which the x-axis represents an SEI indicator, and the y-axis corresponds to an EI indicator. SEI indicators can be composite indices measuring the socioeconomic outcomes derived from the services provided by the geopolitical unit's local SEI. These composite socioeconomic indices generally range from 0 to 1 after normalization and aggregation, with 0 representing the worst and 1 representing the best condition. On the other hand, EI indicators measure if the geopolitical unit is indeed conserving its CNC. EI indicators can be positive (the geopolitical unit conserves more than required to maintain its CNC), zero (the geopolitical unit conserves only what is needed to maintain its CNC), or negative (the geopolitical unit conserves less than required to maintain its CNC).

The location of a geopolitical unit at a given time (given by its x and y values) in the Cartesian plane represents its development state. If at least one of the indicators changes, the geopolitical unit shifts to another development state. A geopolitical unit's successive development states represent its development pathway or trajectory. To provide a normative direction for place-based sustainable development efforts and thus respond to the question of what a sustainable society looks like, Silva and Topf (2020) suggest that the space within the Cartesian plane where societies simultaneously have zero or positive EI scores and high SEI scores (usually >0.8) be termed the "Brundtland Quadrat." Hence, geopolitical units within the Brundtland Quadrat are in sustainable states, whereas the ones outside it are in unsustainable states (Silva and Topf, 2020).

Our approach to quantifying the sustainability of development pathways consists of defining an ideal sustainable development pathway for each geopolitical unit from its initial state (therefore, we assume that development is path-dependent), and then using the Euclidean norm to create an index that measures the distance between the ideal sustainable development pathway and the actual development pathway (both represented as vectors in $\mathbb{R}^2$) in a given time frame (Fig. 1).

Let t_0 and t_1, with $t_0 < t_1$, be two time frames. Assume that the geopolitical unit A has SEI x_0 at t_0 and x_1 at t_1. Also, assume it has EI y_0 at t_0 and y_1 at t_1. We define the pathway vector, v, as the difference:

$$v = (x_1 - x_0, y_1 - y_0) \tag{1}$$

Now, let α be the minimum proportion of land (or sea) required to maintain A's CNC. Thus, the maximum possible EI for A is one minus the minimum proportion of land (or sea) to be protected within the geopolitical unit, or, in mathematical terms, 1 – α. On the other hand, the minimum EI is −α. Assuming that the maximum value of the SEI score is one and the minimum is 0, we define the ideal pathway vector, w, of A starting at (x_0,y_0) as:

$$w = (1 - x_0, 1 - \alpha - y_0) \tag{2}$$

To define the Sustainable Development Pathway Index (SDPI), we first need to determine the SDPI's modulus. Thus, we take:

$$|SDPI(A)| = \frac{100}{\sqrt{2}}||v - w|| = \frac{100}{\sqrt{2}}\sqrt{(x_1 - 1)^2 + (y_1 + a - 1)^2} \tag{3}$$

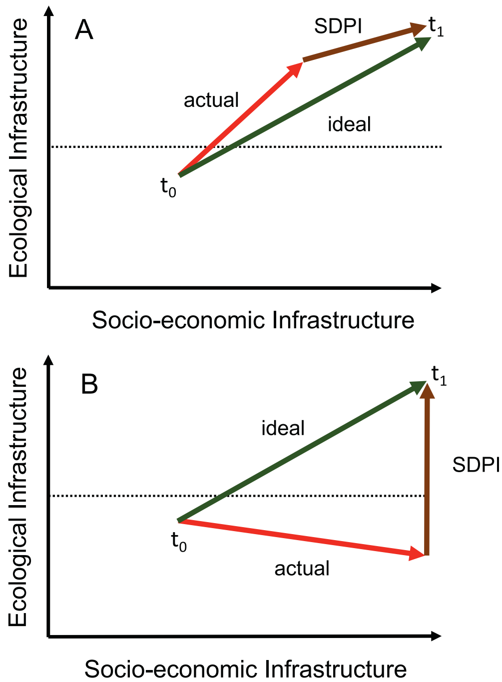

Fig. 1. The Sustainable Development Pathway Index (SDPI) measures the distance between the actual and ideal development pathways. In the example, two geopolitical units are in the same states in t_0. For geopolitical unit A, the SDPI is positive, indicating a sustainable development pathway. On the other hand, for geopolitical unit B, the SDPI is negative, indicating an unsustainable development pathway. The dotted lines represent the critical natural capital.

In other words, |SDPI(A)| is a constant multiplied by the Euclidean distance between the actual development pathway, v, and the ideal development pathway, w. This value may be interpreted as how much A moved away from its ideal sustainable development pathway during the time frame. To facilitate the interpretation of the index, we used the term $100/\sqrt{2}$ in (3) to guarantee that |SDPI(A)| varies between 0 and 100. Therefore, the lower the modular value, the closer the geopolitical unit is to its ideal sustainable development pathway.

However, the SDPI's modulus alone cannot determine if two or more development pathways are sustainable. For instance, geopolitical units in an unsustainable state can reduce their distance to the Brundtland Quadrat by improving their SEI indicators despite maintaining negative EI scores. Thus, we must distinguish between sustainable and unsustainable pathways using normative criteria. We consider a pathway vector, v, to be unsustainable if it satisfies one of the following conditions:

$$y_1 < 0 \text{ and } y_0 > 0$$

$$y_1 \leq y_0 \text{ and } y_0 < 0$$

Hence, v is unsustainable if A either moves from a positive EI value at t_0 to a negative EI value at t_1, or if A has a negative EI value at t_0 and this value does not increase at t_1.

Therefore, we define SDPI(A) as:

$$SDPI(A) = |SDPI(A)| \text{ if v is sustainable}$$

or

$$SDPI(A) = -|SDPI(A)| \text{ if v is unsustainable}$$

Interpreting the SDPI requires using both the sign and modular value. The sign informs if a society's pathway is sustainable or unsustainable, whereas the modular index value represents how close the society is to its ideal pathway. For instance, let us compare two different geopolitical units, A and B. If both SDPI(A) and SDPI(B) have the same sign, the one having the modular value closer to zero is more sustainable because it is closer to the ideal sustainable development pathway. On the other hand, if they have different signs but the same or different modular values, the index with the positive sign represents the most sustainable development pathway because, based on the information currently available (e.g., Dodds et al., 2008), conserving existing ecosystems is always a much more efficient and cheaper development strategy than restoring degraded ecosystems. Finally, although we used a straight line to estimate the ideal sustainable development pathway (i.e., the shortest path between the initial state and the point where EI and SEI are 1), we are not assuming that development pathways are always linear. In fact, if high-resolution temporal datasets representing SEI and EI indicators become available, the SDPI can be estimated over different time frames; thus, its temporal trend can be documented and compared with different mathematical functions to capture the path-dependency and non-linearity of development pathways.

2.2. *Study area*

We used the SDPI to evaluate the sustainability of the development pathways of the 517 municipalities of the Brazilian Amazon between 1991 and 2010. This region is defined based on the limits of the Amazonia Biome as determined by the Brazilian Institute of Geography and Statistics (Instituto Brasileiro de Geografia e Estatística, 2004). These geographical limits generally follow the original extension of the tropical rainforests in northern Brazil. Six states (Amazonas, Acre, Rondônia, Roraima, Amapá, and Pará) have their entire territories within the region (Fig. 2). On the other hand, three states (Mato Grosso, Maranhão, and Tocantins) only partially overlap with the region's limits (Fig. 2). The region covers an area of approximately 4.3 million km^2 and was home to around 19.8 million people in 2010, with most people living in urban centers.

We chose municipalities as the unit of analysis because they are the region's smallest geopolitical units with administrative autonomy and thus represent the local political level. Municipalities are governed by mayors and legislative bodies elected every four years that manage their budgets and establish local regulations and priorities aligned with state and national policies (Silva and Prasad, 2019). Under the triune federalist system of Brazil, the success or failure of initiatives promoted by the federal government depends on the adhesion of municipalities at the local level, as municipalities can choose to either allocate resources in support of those initiatives or choose to not subscribe to them (Neves, 2012). Thus, municipalities can be considered the most fundamental geopolitical units in which studies on development pathways can be carried out across the region.

2.3. *Infrastructure indicators*

We gathered information about EI and SEI indicators for all municipalities in the Brazilian Amazon in 1991 and 2010. To assess SEI quality across municipalities, we used the Municipal Human Development Index (MHDI) as calculated by the Brazilian Development Atlas (www.Atlas http://desenvolvimento.org.br). Although MHDI does not directly measure a municipality's SEI stocks, it synthesizes the outcomes of three essential services provided by SEI: income, health, and education. MHDI ranges from 0 to 1, with 0 representing the worst and 1 representing the best condition. We classified an MHDI value $\geq$0.8 as the "very high" SEI score needed to reach the Brundtland Quadrat (Fig. 3), according to the 0.8 threshold of the "very high" human development category used by the United Nations Development Programme (UNDP, 2020) and Brazilian Development Atlas.

To assess EI quality across municipalities, we used the difference between the proportion of the municipality covered by native vegetation and the proportion of the municipality that should be allocated to conserve its CNC. We used the data produced by the Brazilian Annual

Fig. 2. The political boundaries of the nine states and 517 municipalities of the Brazilian Amazon.

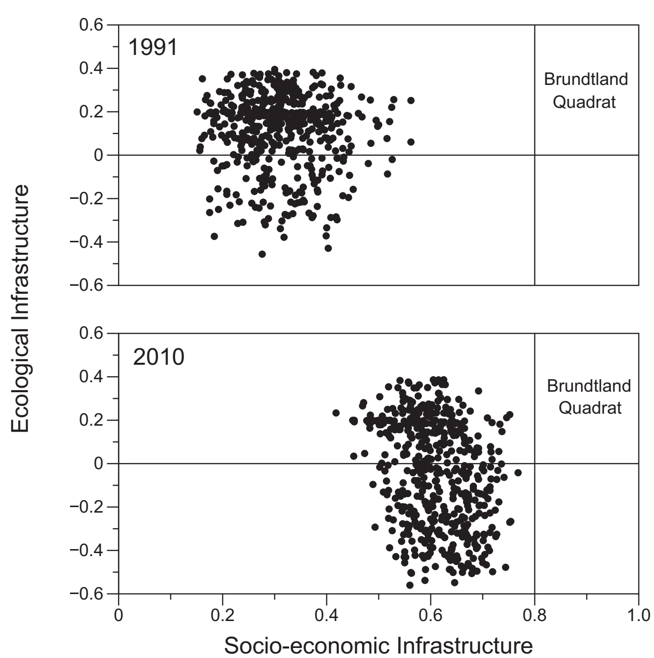

Fig. 3. Changes in the development pathways of the 517 municipalities of the Brazilian Amazon between 1991 and 2010.

Land Use and Land Cover Mapping Project (http://www.mapbiomas.org) to estimate the proportion of a municipality area covered by native vegetation. To define each municipality's CNC, we first grouped the municipalities into quartiles according to their size. For the bottom quartile municipalities (up to 1.07 thousand km^2), we used 0.6 as the CNC. For the top quartile municipalities (>8.19 thousand km^2), the CNC was set at 0.8. For the municipalities in the second and third quartiles, CNC was estimated using the following formula:

$$CNC = (0.8\text{–}0.6)(x - 1.07)/(8.19\text{–}1.07) + 0.6.$$

Where x is the municipality's size (in thousands of km^2). Ideally, the CNC should be discovered through a context-specific process of participatory conservation planning that includes scientists, citizens, and policymakers (Lacher, 2017; Watson et al., 2011). However, using 0.6 and 0.8 as CNC local thresholds in the Brazilian Amazon context is justified because they represent the best estimates of how much native vegetation needs to be maintained to conserve the region's biological integrity and avoid substantial ecosystem service loss (Lovejoy and Nobre, 2019; Ministério do Meio Ambiente, 2007). We classified an EI score ≥ 0 as the value needed to reach the Brundtland Quadrat (Fig. 3) according to the framework modified by Silva and Topf (2020).

3. Results

From 1991 to 2010, all municipalities in the Brazilian Amazon changed their development states, but none reached the Brundtland Quadrat (Fig. 3). Most of the region's municipalities improved their SEIs, generally at the expense of their EIs (Fig. 3). Because of such changes, the distribution of the SDPI is strongly bimodal, indicating the existence of two distinct clusters of local development pathways across the region separated by a large gap between them (Fig. 4). The first cluster comprises 246 municipalities that followed sustainable development pathways, whereas the second cluster comprises 271 municipalities that followed unsustainable pathways (Table 1). The first cluster of municipalities encompasses 3.3 million km^2 and was home to 11 million people in 2010. On the other hand, the second cluster covers 1.0 million km^2 and was home to 8.7 million people in 2010 (Table 1). Therefore, although 52.4% of municipalities followed unsustainable pathways, they represent only 23% of the region's area and 44% of its population.

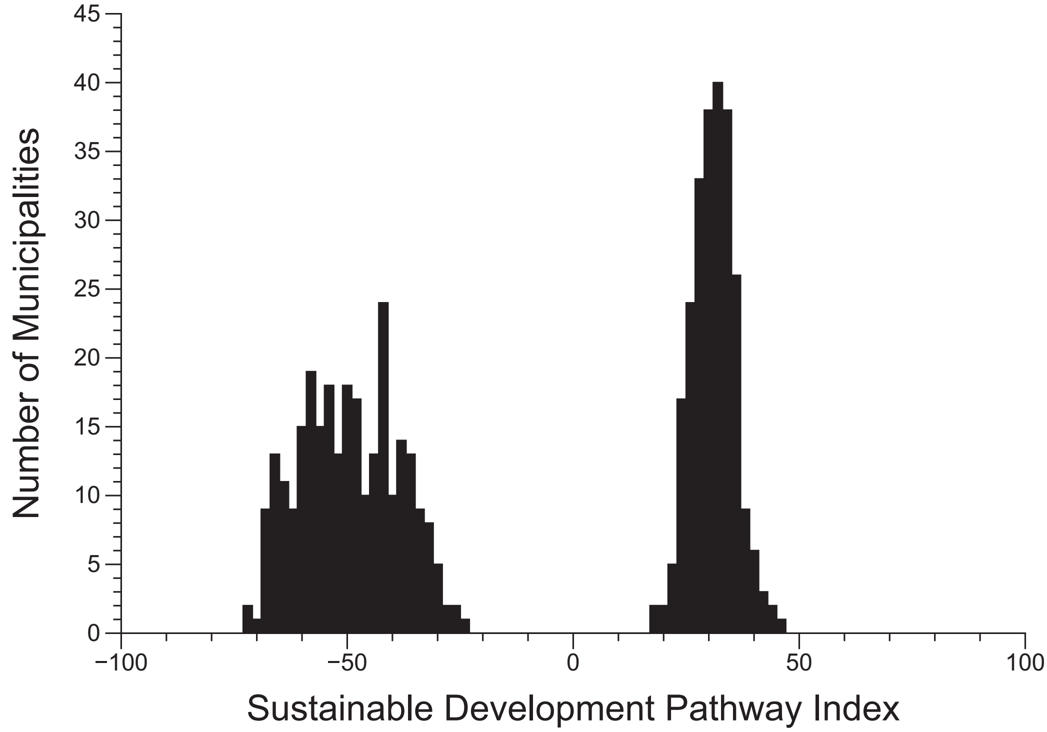

Fig. 4. Municipalities of the Brazilian Amazon clustered into two groups based on their Sustainable Development Pathway Indices.

Table 1
Distribution of Brazilian Amazon municipalities by state and development pathway category. Development pathways correspond to the period between 1991 and 2010.

States	Sustainable			Unsustainable		
	n	Area (km^2) *1000	Population[1]	n	Area (km^2) *1000	Population[a]
Acre	14	140	300,346	8	24	433,213
Amazonas	62	1559	3,483,985	0	0	0
Amapá	16	143	669,526	0	0	0
Maranhão	42	36	917,578	53	83	2,775,633
Mato Grosso	13	171	155,226	70	422	1,079,685
Pará	74	972	4,848,458	69	276	2,732,593
Roraima	7	61	176,059	0	0	0
Rondônia	15	224	450,479	45	176	1,361,958
Tocantins	3	4	23,731	26	31	354,166
Region	246	3310	11,025,388	271	1012	8,737,248

[a] Population in 2010.

The spatial distribution of the two clusters of municipalities is geographically clustered (Fig. 5). While most municipalities following sustainable development pathways are found across the entire region, municipalities following unsustainable development pathways are mostly concentrated along the region's southern and eastern boundaries, along the arc of deforestation (Becker, 2006; Nobre et al., 2016), with few along the main roads built in the region's core (Fig. 5). Within the clusters of municipalities following sustainable and unsustainable pathways, there is a wide variation in the modular values of the SDPI, indicating high heterogeneity in the development pathways across the region (Fig. 4). Municipalities following sustainable development pathways are less spread out around the SDPI mean (mean = 31, SD = 4.73) than the municipalities following unsustainable development pathways (mean = −49.4, SD = 10.9). This difference is statistically significant (variance ratio test, $p < 0.0001$).

The proportion of municipalities following sustainable development pathways varies across the region's nine states. Based on this variation, states can be classified into three groups. The first group comprises three states (Amapá, Amazonas, and Roraima) where all municipalities followed sustainable development pathways (Table 1). The second group includes three states (Mato Grosso, Tocantins, and Rondônia), where the number of municipalities that followed sustainable development pathways is significantly lower (Chi-square tests, df = 1, $P < 0.0001$) than the number of municipalities that followed unsustainable development pathways (Table 1). Finally, the third group includes three states (Acre, Maranhão, and Pará), where the number of municipalities that followed sustainable development pathways is not statistically different (Chi-square tests, df = 1, $P > 0.05$) from the number of municipalities that followed unsustainable development pathways (Table 1).

4. Discussion

4.1. Patterns of variation in development pathways in the Brazilian Amazon

The SDPI identified a wide variation in the development pathways across the 517 municipalities of the Brazilian Amazon from 1991 to 2010. Such variation is due to two components. The first component, indicated by the SDPI sign, sets municipalities with sustainable development pathways apart from municipalities with unsustainable development pathways. The second component, measured by the SDPI modular value, shows how far the municipalities are from their ideal development pathways. Altogether, the SDPI's two components showed that the Brazilian Amazon is a heterogenous region with development pathways varying widely at the local level.

The three groups of states recognized in the SDPI analysis align well with the three historic stages of expansion of economic frontier development from the Brazilian Amazon's southern and eastern borders to its center (Becker, 2006). The states dominated by unsustainable development pathways compose the consolidated economic frontier, which is characterized by the adoption of the conventional development model

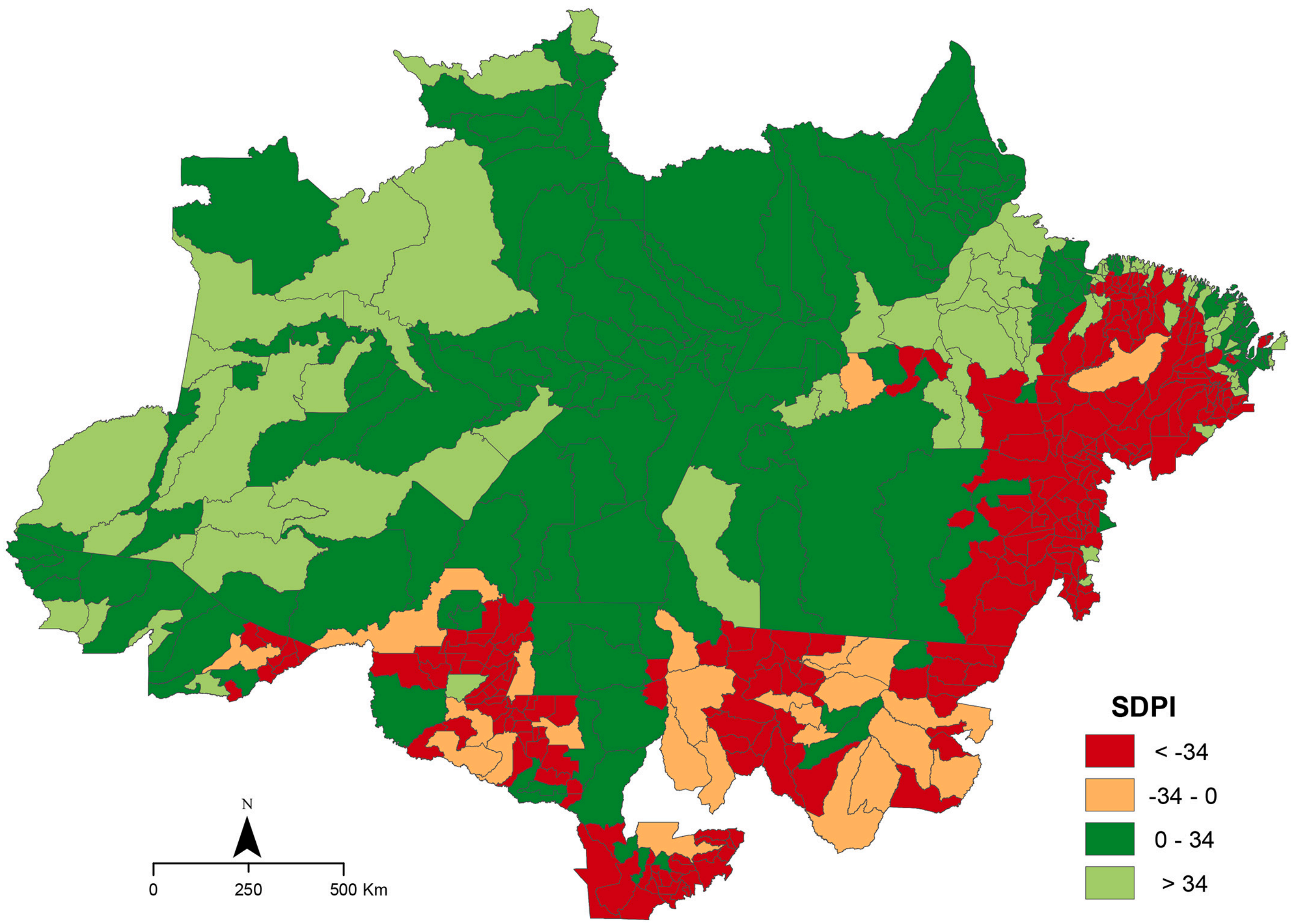

Fig. 5. Spatial distribution of sustainable (SDPI >0) and unsustainable (SDPI <0) development pathways across the 517 municipalities of the Brazilian Amazon between 1991 and 2010.

that seeks to generate endless economic growth based on the unsustainable use of natural resources (Becker, 2006; Garda et al., 2010; Silva et al., 2017). These southern and eastern states have experienced increased deforestation since the 1960s due to expanding infrastructure projects and agricultural activities (Fearnside, 2005; Mahar, 1979). In this group, sustainable development can only be achieved through the accelerated restoration of EI associated with improvements in SEI through economic activities that efficiently use cleared lands. On the other extreme, there is the sustainable development frontier in the north and west, composed of states in which all municipalities followed sustainable development pathways, possibly because of the protected areas, indigenous lands, and social programs established by the federal and state governments to ensure simultaneously long-term nature conservation and poverty reduction (Caviglia-Harris et al., 2016; Silva et al., 2017). Among this group, Amapá was the only state in the Brazilian Amazon that adopted sustainable development policies throughout most of the study period, regardless of the political party in power (Cunha et al., 2019). In these states, achieving sustainable development requires maintaining EI while improving SEI via support for forest-friendly economic activities (Garda et al., 2010; Nobre et al., 2016). Finally, between these two extremes, there is the modern economic frontier composed of states with relatively equal numbers of municipalities following sustainable and unsustainable development pathways. In this group, sustainable development can only be achieved by ambitious conservation and restoration of EI plus SEI improvements driven by accelerated adoption of forest-friendly economic activities.

These three general patterns of state-level development and the recommendations derived from them are consistent with the fact that the Brazilian Amazon is a large ecological region that has been passing through a relatively recent and accelerated process of occupation driven by the expansion of economic activities based on extensive land transformations (Becker, 2016; Becker et al., 1990; Garda et al., 2010; Nobre et al., 2016; Thaler et al., 2019). However, studies show that the magnitude of the consumption of the region's ecosystems far exceeds the magnitude of the expected improvements in human well-being (Caviglia-Harris et al., 2016; Celentano et al., 2012; Silva et al., 2017), possibly because the economic gains generated from consuming the region's EI have not been efficiently reinvested in the region's SEI (Barbier, 2020).

4.2. Is the SDPI sensitive enough to represent place-based development pathways?

The empirical application of the SDPI in the Brazilian Amazon shows that the index described the local development pathways across a large and heterogeneous frontier region well while generating geographic patterns that match the region's recent history. Therefore, it can be considered sensitive enough to local conditions to be used worldwide. This sensitivity to local conditions is relevant to studies on the sustainability of development pathways because most of the studies on this subject have been carried on at the national level (Koehler et al., 2019; Loorbach et al., 2017) and fail to assess how local biophysical, historical, and socioeconomic factors shape development pathways within countries or large regions (Silva et al., 2020). Moreover, at the national level, development pathways tend to be generally unsustainable (e.g., Carrasco et al., 2017; O'Neill et al., 2018). However, our exploratory study shows a much more complex and intriguing pattern at the local level, suggesting that modern societies can be better described as a complex mosaic of interacting local socio-ecological systems following different development pathways. The sustainability or unsustainability of these local development pathways is, in turn, determined by the interaction of internal and external factors (including, for instance, the interactions with other geopolitical units). These factors may be geographical, cultural, historical, religious, political, or economic (Barbier, 2016; Cumming and von Cramon-Taubadel, 2018; Silva and Topf, 2020). Understanding the relative contribution of these different factors to local development pathways is an exciting and relevant research agenda that still needs further development. Combining the SDPI and modern geospatial statistical methods, for instance, will allow scholars to identify major global patterns of development pathways and their covariates. Such analyses, in turn, should be followed by in-depth, mixed methods studies aiming to unfold the processes determining how local societies navigate their transitions toward or away from their ideal sustainable development pathways.

4.3. Improving EI and SEI indicators

As with any method used to represent complex, multidimensional processes, the SDPI's usefulness is limited by the information used to calculate it. Fortunately, the SDPI is flexible enough to accommodate context-specific indicators and can be used at any geopolitical scale (Colladós and Duane, 1999; Silva and Topf, 2020). As a starting point for comparative studies within and among countries, we recommend using the Human Development Index (HDI) as a proxy for SEI. The main advantages of HDI over other similar SEI composite indices are that HDI is well known by decision-makers, is focused on three of the essential outcomes generated by the services provided by SEI (health, education, and income generation), and, most importantly, is available at both national (UNDP, 2020) and sub-national levels (Permanyer and Smits, 2020) at relatively good temporal resolutions. Nevertheless, studies should also recognize that HDI has two main limitations when used as an SEI proxy. The first is that HDI does not consider social inequality, an important challenge to be overcome in any sustainable development strategy (UNDP, 2020). The second is that HDI does not measure SEI efficiency in producing gains in well-being over time. SEI efficiency is usually a consequence of governance capacity (Andrews, 2014), which, in turn, is also a critical element of sustainable development (Sachs, 2015). Ideally, a comprehensive SEI indicator should combine the characteristics of the inequality-adjusted HDI proposed by the United Nations Development Programme (UNDP, 2020) with the characteristics of the Sustainable Human Development Index (SHDI) proposed by Assa (2021) that seems to be a robust measure of SEI efficiency.

To evaluate EI gaps, we suggest following two steps. First, use systematic conservation planning to define the extent and the configuration of the EI required to maintain CNC (Silva and Topf, 2020). Then, use natural or restored ecosystem maps to evaluate how well the actual EI matches the ideal EI. Systematic conservation planning is a multidisciplinary and multi-stage process that combines scientific data and analysis with a participatory approach to plan optimal EIs based on the spatial distribution of ecological assets worth being protected (Lacher, 2017; Margules and Pressey, 2000; Watson et al., 2011). Originally, systematic conservation planning was focused solely on species distribution and ecosystems (Margules and Pressey, 2000; Rodrigues et al., 2004). However, recent studies include ecosystem services (Kukkala and Moilanen, 2017; Liquete et al., 2015; Maes et al., 2015), in addition to considering conservation costs (Naidoo et al., 2006) and climate change scenarios (Maxwell et al., 2019; Reside et al., 2018). In the past, the use of systematic conservation planning was limited because of computational processing capacity and data availability, but these shortcomings have been reduced in the last two decades (Lacher, 2017; Silva and Topf, 2020) and CNC national-level estimates are available for several countries (e.g., Allan et al., 2022). Therefore, from a practical perspective, assessing and monitoring a society's EI using modern technologies while fostering an inclusive, participatory approach is a task as tangible as evaluating and monitoring social and economic indicators.

5. Conclusion

The SDPI sets the standards of what a sustainable society looks like and distinguishes between sustainable and unsustainable development pathways; thus, it fills an important gap in the existing set of metrics proposed to assess place-based sustainability transitions. Because the SDPI shows how far a society is from its ideal sustainable development pathway based on its current state, the index can be used by decision-makers to define or prioritize policies and programs needed for reorienting place-based development pathways toward a sustainable state (Beer et al., 2020). Reorienting development pathways can combine technology-led, market-led, state-led, and citizen-led processes (Scoones, 2016). In addition, these processes can be adaptative if they improve current conditions without creating a rupture with the dominant socioeconomic system (Mol et al., 2013); transformative if they enhance current conditions by promoting substantial changes in the dominant socioeconomic system (Curran, 2017; Herrfahrdt-Pahle et al., 2020); or adaptative-transformative if they combine both adaptive and transformative approaches (Pant et al., 2015). Regardless of the combination of processes used to accelerate the transition to a more sustainable state and the magnitude of the changes required to increase their SDPIs, societies must have (a) an enabling external environment; (b) internal capacity to create and implement systemic changes; and (c) enough public support to overcome challenges and maintain the social and ecological gains achieved during their sustainability journeys.

Funding sources

This work was supported by funding from the University of Miami and the Swift Action Fund.

Data availability

No new data were created or analyzed for this article. All information is publicly available.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We are grateful to Luis Barbosa, who created all of the maps, and two anonymous reviewers for constructive comments on a previous version of this paper.

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